DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Single and Multivariate Calculus | Course Code: 25EN1201 | Credits: 4

Exam Session: Mid Semester Exam (CIA-1) March 2026 (Max Marks: 40, Duration: 75m) Max Marks: 80 | Duration: 3 Hours

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INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Find the maximum and minimum values of the function $f(x, y) = 2x + y$ subject to the constraint $x^2 + y^2 = 5$ using the method of Lagrange Multipliers. **[8 Marks]**

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Q2. Verify Green's Theorem in the plane for the vector field $F = (xy + y^2)\, i + x^2\, j$ around the closed curve bounded by $y = x$ and $y = x^2$. **[8 Marks]**

MODULE 2

Q3. Apply Gauss Divergence Theorem to compute the flux of $F = 4xz\, i - y^2\, j + yz\, k$ across the surface bounded by the cylinder $x^2 + y^2 = 4$ and planes $z = 0$, $z = 3$. **[8 Marks]**

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Q4. Test the convergence of the infinite series sum from n=1 to inf of $[(n!)^2 / (2n)!] * x^n$ ($x > 0$) using D'Alembert's Ratio Test. **[6 Marks]**